

Gene Trigo

Senior Software Engineer — Gameplay & Engine Systems

Unreal Engine 5 · C++ · Unity · C# · Applied Mathematics

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México · Open to remote

ENGINES

Unreal Engine 5 / 4 · Unity · Kanzi · Godot · Havok · Custom C++ engines

LANGUAGES

C++ · C# · Blueprints · HLSL / GLSL · Objective-C · Swift

SPECIALTIES

- Gameplay systems & GAS
- Multiplayer / networking
- Performance optimization
- Shaders & rendering (OpenGL)
- Procedural geometry & mesh algorithms
- Editor & debug tooling
- Cross-platform porting
- Simulation & numerical methods

TOOLS

JetBrains Rider · Visual Studio · Git · FMOD · Photon Fusion 2 · Azure · PlayFab · Photoshop

EDUCATION

B.Sc. Mathematics & Physics
Major in Applied Mathematics for Computer Science
[University] · 2007 – 2015

SPOKEN LANGUAGES

Spanish — Native
English — Professional working

PROFILE

Senior software engineer with 15+ years writing C++ and 10+ years building games across Unreal Engine, Unity and custom engines — from AAA mobile ports at Gameloft to UE5 gameplay, multiplayer and systems work for PC and consoles. Trained as a mathematician and physicist, I bring that rigor to simulation, rendering, optimization and tooling, and I enjoy mentoring teams toward the right technical path.

EXPERIENCE

Senior Engineer · Titan Games — 2022 – Present

Engineering partner embedded with game studios on client projects · Remote

- Designed and built the complete weather system for a UE5 arcade flight simulator on PC and consoles: wind simulation, clouds, FMOD weather audio, editor preview and QA debug tools.
- Implemented the Gameplay Ability System and drove performance optimization for a UE5 multiplayer shooter.
- Developed an LSCM mesh-unfolding plugin for UE4 and procedural mesh generation for a 3D design application.
- Built Unity multiplayer on Photon Fusion 2 with Azure + PlayFab; lead developer on several casual and mobile titles.

Senior Engineer · Visteon — 2018 – 2022

Automotive cockpit graphics · Kanzi, Godot, Unity

- Re-engineered existing shaders to run on low-end GPUs without reducing visual quality.
- Simulated 3D lighting, depth and roughness on 2D nodes with OpenGL; built a procedural highway with generated traffic.
- Guided teams across projects on technical direction; delivered trainings and video documentation of recurring issues.

Engineer · Gameloft — 2015 – 2018

Modern Combat Versus · Gangstar New Orleans · Siegefall

- Ported shipped titles from iOS to Android, Windows Phone and Windows Store in custom C++/C# engines.
- Owned the input system on Modern Combat Versus; made Havok compatible with Windows Store for Gangstar New Orleans.
- Investigated cross-platform multiplayer desyncs caused by floating-point differences on Siegefall.

Earlier roles

Temple · Mathematician Programmer & iOS Developer — subsoil time-series tool, iOS HR app — 2013 – 2014

Nemfis Applied Science · Mathematician Programmer — linear-algebra solvers, pathfinding, rendering — 2009 – 2012

Arcade Flight Simulator

UE5 · C++ · PC & consoles · Titan Games

Designed and implemented the weather system end to end: wind simulation, cloud rendering, weather-driven FMOD audio, an in-editor preview for designers, and debug tooling for QA.

Star Plunder

UE5 · C++ · Multiplayer shooter · Titan Games

Gameplay-focused role. Implemented abilities on Unreal's Gameplay Ability System and led performance optimization work across the game.

3D Mesh Unfolding Plugin

UE4 · C++ · LSCM · Titan Games

Built a UE4 plugin implementing Least Squares Conformal Maps to unfold 3D models into flat patterns for a third-party design application, alongside procedural mesh generation.

Beanland

Unity · C# · Photon Fusion 2 · Azure · PlayFab

Online multiplayer game built on Photon Fusion 2 networking, with Azure and PlayFab backend services.

Koi Pond & casual titles

Unity · C# · Mobile · Lead developer

Built Koi Pond's grid system and math-heavy gameplay logic. Led development on casual and hyper-casual titles: drag & drop, arcade, drop puzzle, sudoku-style and merge games.

Automotive cockpit graphics

Kanzi · Godot · Unity · OpenGL · Visteon

Faked 3D lighting, depth and roughness on 2D nodes; optimized shaders for low-end GPUs at unchanged quality; generated a procedural highway with traffic.

Modern Combat Versus

Custom engine · C++ / C# · Gameloft

Member of the Android porting team; responsible for the input system of this competitive online FPS.

Gangstar New Orleans

Custom engine · Havok · Android · Gameloft

Made Havok compatible with Windows Store builds; fixed 3D rendering bugs and optimized performance for the Android port.

Siegefall

Custom engine · C++ / C# · Gameloft

Researched multiplayer desync errors caused by floating-point behavior across platforms; general bug fixing.

Earthquake Impact Simulator

C++ · 3D rendering · Scientific computing

Programmed the seismic-wave impact simulation and 3D visualization for the Valley of Mexico, working with a physics PhD on paper analysis and research.

IEDF 2012 Election Forecasting

C++ · Statistics · Mexico City electoral institute

Built preliminary-results and outcome-prediction software for the 2012 Mexico City election, implementing the "Conteos Dinámicos" paper's algorithm.

MAAK

iOS · Swift · Objective-C

Full-stack iOS development: built both the app front end and its back end.